

HRITAM SARKAR

Electronics and Communication Engineer

 Siliguri, West Bengal, India

 +91-6296429319

 hritamsarkar@gmail.com

 Github: <https://github.com/Hoptimus>

PROFESSIONAL SUMMARY

Electronics and Communication Engineer experienced in embedded systems, hardware, and networking. Skilled in implementing innovative ideas in electronics and software. Pursued an MCA with a focus on AI/ML, microcontroller interfacing and intelligent system design. Recognised for analytical thinking, adaptability, and solution-orientated execution.

TECHNICAL SKILLS

- **Programming Languages:** C, Java, Python, PHP, JavaScript, HTML, CSS, R, MATLAB
- **AI Agent Platform:** Claude Code, Antigravity IDE, MCP Server
- **Embedded & Hardware:** Microcontroller Programming, Circuit Design, PCB Design, Computer Assembly & Maintenance
- **Software & Tools:** Microsoft Office Suite, Photoshop, Canva, Audio & Video Editing Tools, Scilab, Octave
- **Signal And System:** Object Tracking and Decoding Signal Coming from Object.
- **Additional Skills:** Project Management, Responsive Web Design, Remote Sensing (ISRO Certified)

PROFESSIONAL EXPERIENCE

Computer Hardware & Network Engineer

Kore Infosystem, Kolkata | Aug 2022 – Oct 2023

- Diagnosed and repaired computer hardware issues across diverse client systems
- Configured and maintained LAN/WAN networks
- CCTV camera configuration with NVR/DVR and Wi-Fi Camera setup

Computer Hardware Engineer

Mobile Computer Repair Center, Siliguri | Nov 2019 – May 2022

- Performed mobile and desktop repair, upgrades, and optimization
- Built long-term client trust through reliable technical service

Operation Maintenance Control Remote Executive

Technocon Services, Siliguri | Jul 2019 – Nov 2019

- Telephone and mobile tower monitored and controlled remote operations and maintenance systems
- Create management system with technician and higher authorities to maintain the phone tower

EDUCATION

Master of Computer Application (MCA)

University of North Bengal | Oct 2023 – 2026

- Major Focus: Computer Programming, Artificial Intelligence, Machine Learning
- Research and study at High Energy and Cosmic Ray Research Centre

Bachelor of Technology (B.Tech) – Electronics & Communication Engineering

MAKAUT, West Bengal | Aug 2014 – Aug 2018

- Relevant Coursework: Electronics, Communication Systems

KEY PROJECTS

HUNTER-K1 (MCA Final Year Project)

- Wheel-based robotic arm with computer vision, object detection and AI integration

Gesture-Controlled Robotic Arm (B.Tech Final Year Project)

- Analog gesture control using LDR matrix

Other Projects

- Variable Speed Digital Clock using 555 Timer, Object Counter System
- Running Off-line and Off-grid messaging platform for a new era of secure communications (Open Source).
- Binary detection of eczema vs non-eczema using a Convolutional Neural Network (CNN).
- Binary detection of Gender Classification (Male and Female) using a Convolutional Neural Network (CNN).
- Real-time Binary detection (Yolo) of Gender Classification (Male and Female) using a CCTV
- Industrial Multi-Agent AI & Knowledge Augmented Generation (KAG) system that utilizes the Model Context Protocol (MCP) to manage a local AI environment .
- Tracking Satellite and Decoding Signal Coming from Satellite for Educational Purpose.

PUBLICATIONS

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| • “Home-Based and Portable Smart Weather Station” | IISRR (ISSN: 2394-885X) |
| • “Real-Time Monitoring of Radiant Flux and Surface Thermal Properties via Arduino Thermocouple” | IJSR (ISSN: 2319-7064) |
| • “Real-Time Wireless Sound Localisation and Frequency Analysis: A Low-Latency IoT Approach” | IJSR (ISSN: 2319-7064) |
| • “Real-Time Weather Monitoring Using NodeMCU for Diverse Application” | IIP (ISSN: 3070-7617) |

CERTIFICATIONS & TRAINING

- Remote Sensing and Digital Image Analysis (ISRO).
- Training on Radio Broadcasting and AM,FM transmission at All India Radio (AKASHVANI).

WEBSITE

- Solaris Space Weather OPS - Multi-Agent AI Based Space Weather Monitoring System